Final examination on Prolog

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1. Let `delete(X,S,T)` be a relation true when the list `T` contains the same items as the list `S`, in the same order, except the first `X` in `S` (starting from the top). One possible definition is

   ```
   delete(X,[X|A],A).                          % Rule 2
   delete(X,[Y|A],[Y,B]) :- delete(X,A,B).     % Rule 4
   ```

   We want to modify this definition so that the relation is true when `S` does not contain any `X`.

   Question. What happens if we add the fact

   ```
   delete(_,X,X).                              % Rule 1, 3 or 5?
   ```

   as rule 1, 3 or 5? Give counter-examples if needed.

2. **Question.** Define a relation `catenate(U,V,W)` to be true when list `W` is made of list `U` followed by list `V`. For example

   ```
   ?- catenate(U,[3,4],[0,1,2,3,4]).
   U = [0,1,2] ;
   No
   ```

3. **Question.** Define a relation `flatten(A,B)` to be true when `B` is the list of the items found in the list of lists `A`. (**Hint.** Use `catenate`.)

   ```
   ?- flatten([[3,-1,-1],[0],[]],B).
   B = [3,-1,-1,0] ;
   No
   ```

4. **Question.** Define a relation `split(L,P,N)` to be true if `P` contains the positive or nul items of `L` (same order as in `L`) and `N` contains the negative items (same order). For example:

   ```
   ?- split(L,[1,0],[-3,-2]).
   L = [1, 0, -3, -2] ;
   L = [1, -3, 0, -2] ;
   L = [1, -3, -2, 0] ;
   L = [-3, 1, 0, -2] ;
   L = [-3, 1, -2, 0] ;
   L = [-3, -2, 1, 0] ;
   No
   ```